

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

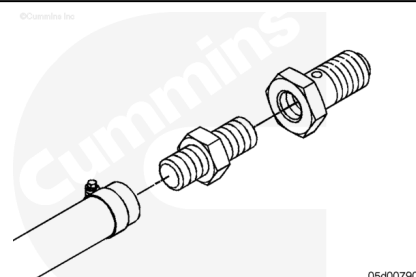
- Electrical contact cleaner, Part Number 3824510
- Fuel System Clean Care Kit, Part Number 4919073.

Additional Service Items

- No additional service items required.

Initial Check

Use the following procedure for instructions to measure the fuel pump head drain. Refer to Procedure 005-236 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-236.html>)



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system or internal engine components to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component

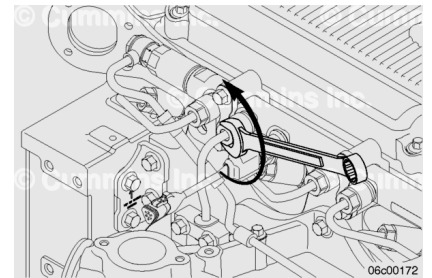
to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine.

Clean the fuel pump head and surrounding area with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Normal engine operation creates highly pressurized fuel in the fuel line which will remain in the fuel line after engine shutdown. Never open the fuel system when the engine is operating. Before servicing the fuel system, always loosen the pump-to-rail fuel line at the rail to vent the pressure. Keep hands clear of the line when loosening. High pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



Before servicing the fuel system, loosen the pump-to-rail line at the rail to vent the pressure.

When loosening the nut, keep hands clear of the line.

Tighten the fuel rail nut.

Torque Value: 47 n·m [35 ft-lb]

Note : A machined slot in this fitting directs the fuel spray toward the engine block.

- Disconnect the high-pressure pump-to-rail fuel line from the high-pressure fuel pump head. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- Remove the fuel drain connection from the fuel pump actuator housing. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-013.html>)

Note : The fuel line from the ECM cooling plate to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. The screen must be examined and cleaned, or the banjo bolt replaced if the screen is damaged.

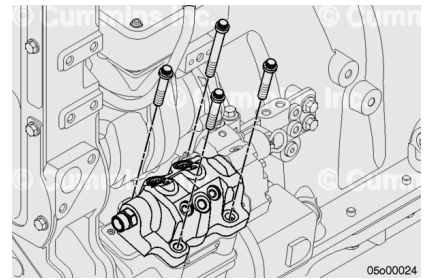
- Remove the fuel supply line from the fuel pump actuator housing. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)
- Disconnect the wiring harness from the fuel pump actuator.

Remove

⚠ CAUTION ⚠

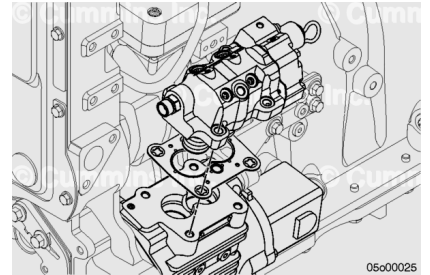
Do not use an air tool. The use of air tools can possibly damage the fuel pump.

Remove the four capscrews that attach the fuel pump head to the fuel pump camshaft housing.



Carefully lift the fuel pump head and gasket from the fuel pump camshaft housing. Be careful to keep the tappet springs attached to the fuel pump head. Place the fuel pump head on a clean surface.

Remove the fuel pump actuator housing. Refer to Procedure 005-228 in Section 5.
(/qs3/pubsys2/xml/en/procedures/377/377-005-228.html)



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

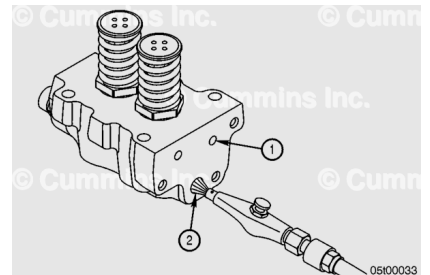
Special care **must** be taken to be sure these parts are kept extremely clean if removed. Cover the camshaft housing with a clean shop towel while the head is removed.

Do **not** use cleaning agents, other than contact cleaner, on pump components.

Note : Do not perform the following test without the springs and spring retainers installed.

With the springs and spring retainers installed, use a rubber tipped air gun, place the nozzle into the inlet (2) and blow compressed air (at least 276 kPa [40 psi]) into the inlet (2).

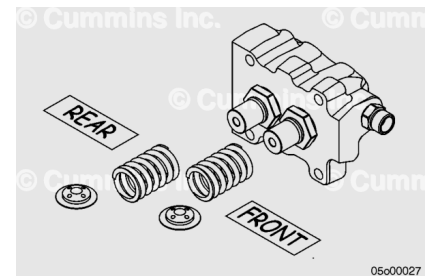
Callout (1) is the outlet.



The plungers should extend. If the pumping plungers are stuck, replace the fuel pump head.

Remove the springs and spring retainers from the barrel retainers.

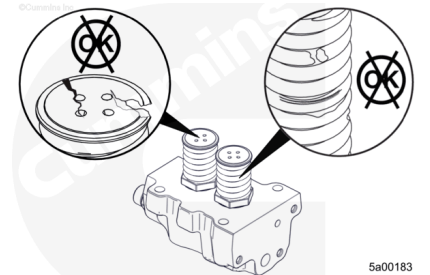
Make certain to keep track of which spring came from the front and rear. It is recommended that these parts be installed in the same location, even if a new high-pressure pump head is installed.



Inspect the barrel springs and spring retainers for cracks or significant wear.

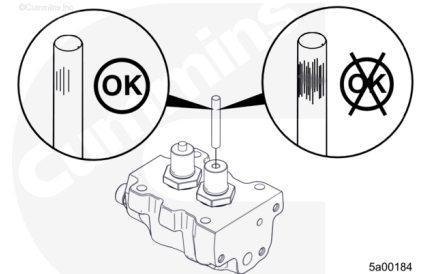
Slight scoring or debris traces on the barrel spring retainer is acceptable.

If cracks, nicks, or debris generated malfunctions are present, replace the fuel pump head.



⚠ CAUTION ⚠

Each plunger must be installed in the same orientation and in the same barrel, or engine damage can result. Marking the bottoms of the plungers with a felt tip marker will help to make sure correct orientation is maintained.



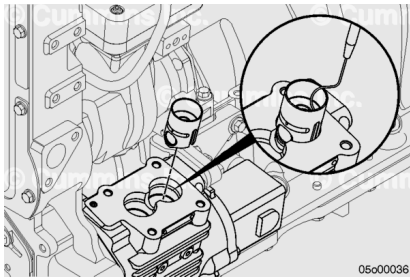
For steel fuel pump plungers, slight discoloration or debris traces are acceptable. Deep scoring or dark discoloration should **not** be evident. If scoring or scratches exist that can be felt, replace the fuel pump head barrel.

If the ceramic plungers are shattered, the fuel pump assembly **must** be replaced.

Note : Some fuel pump heads are built with asymmetric pumping plungers. The crowned end of the pumping plunger **must** be installed into the barrel. If the plunger is installed in the wrong orientation, fuel pump head damage will result.

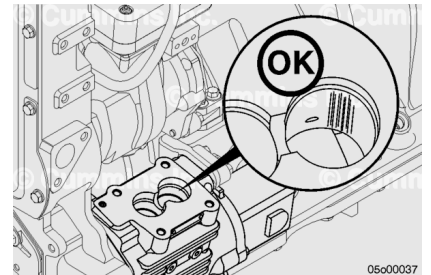
While the fuel pump head is removed, inspect the camshaft housing.

The tappets can be removed. Use an o-ring pick.

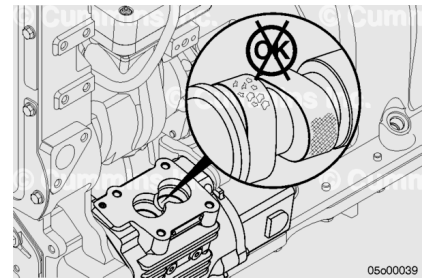


Normal operation creates vertical grooves in the cylinder bores of the fuel pump camshaft housing. These grooves are **not** an indication of a malfunction.

Camshaft housings with grooves are acceptable for reuse.



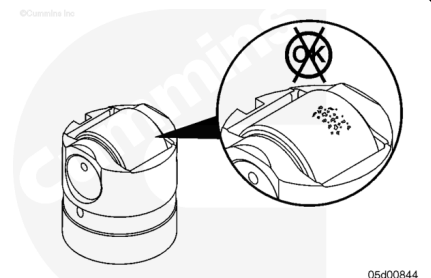
With the camshaft housing tappets removed, inspect the camshaft for wear.



Inspect the tappet rollers for wear.

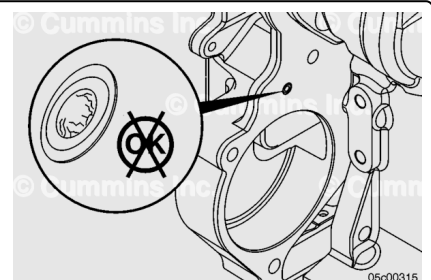
Normal operation creates bands around the circumference of the tappet roller. These bands are **not** an indicator of a malfunction. Tappet rollers with bands are acceptable for reuse.

If damage to the tappet rollers is found, replace the fuel pump assembly.



If damage to the camshaft, tappets, or camshaft housing is observed, it is possible the fuel pump is **not** receiving adequate lubricating oil.

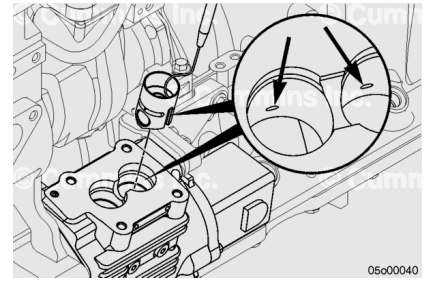
When replacing the fuel pump, inspect the gear housing to make sure no blockages exist in the oil supply to the fuel pump.



Install

⚠ CAUTION ⚠

Each tappet must be installed in the same orientation and in the same barrel from which it was removed. Failure to do so can result in engine damage. Marking the tops of the plungers with a felt tip marker will help to make sure that the correct orientation is maintained.



Install the tappets in the original tappet bores.

Be certain the tappets are installed in the original locations.

Be certain that the tappet guide pins engage the guide pin grooves.

Tappet guide pins **must** be aligned so that the guide pin flats are perpendicular to the fuel pump head mounting surface.

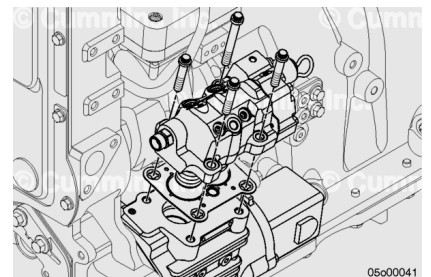
Rotate the guide pin to the appropriate position if the guide pin is misaligned.

If installing a new pump head, install the new tappet springs and retainers provided with the pump head.

Place the high-pressure fuel pump head and gasket onto the high-pressure fuel pump camshaft housing.

Draw the high-pressure fuel pump head down by alternately tightening the four outer high-pressure fuel pump head capscrews until the head just contacts the camshaft housing.

Tighten the high-pressure pump fuel head capscrews to the final torque, starting from the center outward.



Torque Value: 68 n•m [50 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Hold the banjo fittings while tightening the banjo bolt to prevent fitting rotation. Allowing the banjo fitting to rotate may damage the fuel line.

- Install the fuel pump actuator housing. Refer to Procedure 005-228 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-228.html>)
- Connect the fuel supply line to the fuel pump actuator housing. Return the marked banjo bolt to its original location. Failure to do so will result in system contamination. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)

Note : The fuel line from the engine control module cooling plate to the high-pressure fuel pump contains a filter screen at the high-pressure fuel pump banjo fitting. The screen must be examined and cleaned, or the banjo bolt replaced if the screen is damaged.

- Connect the fuel drain connection to the fuel pump actuator housing. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-013.html>)
- Connect the fuel pump-to-rail fuel supply line to the fuel pump. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- Connect the engine harness to the fuel pump actuator.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature. Check for leaks and proper operation.